



PixelProbe – AI Image Detection System

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Abstract

- PixelProbe is a binary classification model designed to distinguish between real and AI-generated images.
- Built on the EfficientNet CNN family.
- Prioritises cross-dataset robustness for high-confidence inference on unseen data.

Motivation

- AI systems produced approx. 34 million images per day in 2023 only a year after the release of ChatGPT with over 15 billion generated by August 2023. [1][2]
- This surge in fake media presents a growing challenge for media verification, digital forensics and public trust online.
- PixelProbe provides a full-stack application that classifies images as either AI-generated or authentic.

Methodology & Datasets

- EfficientNet chosen as base model, pretrained on ImageNet.
- Transfer learning: base frozen, custom classifier head added.
- Pre-processing: resize to 224×224 , normalisation, augmentation (flip, zoom, rotation).
- Datasets Used: 6 datasets used to date (inc. CIFAKE, GENImage, Places365)

Model Training Pipeline

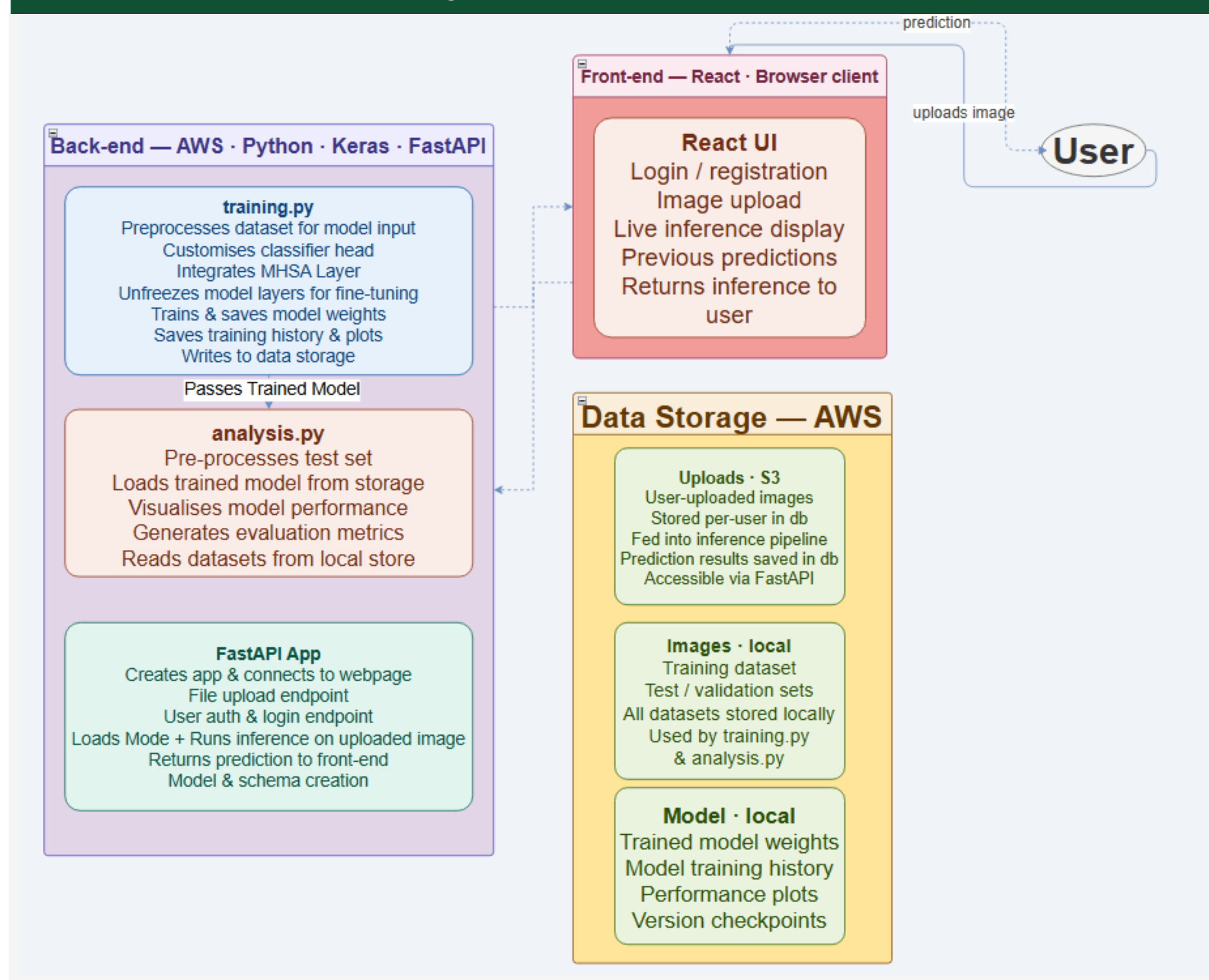
Phase 1: Train classifier head only — stabilises model learning.

Phase 2: Unfreeze top layers and fine-tune with a low learning rate.

Loss: Binary cross-entropy

Early stopping and model checkpoint callbacks ensure best model retention.

System Architecture



Web Application

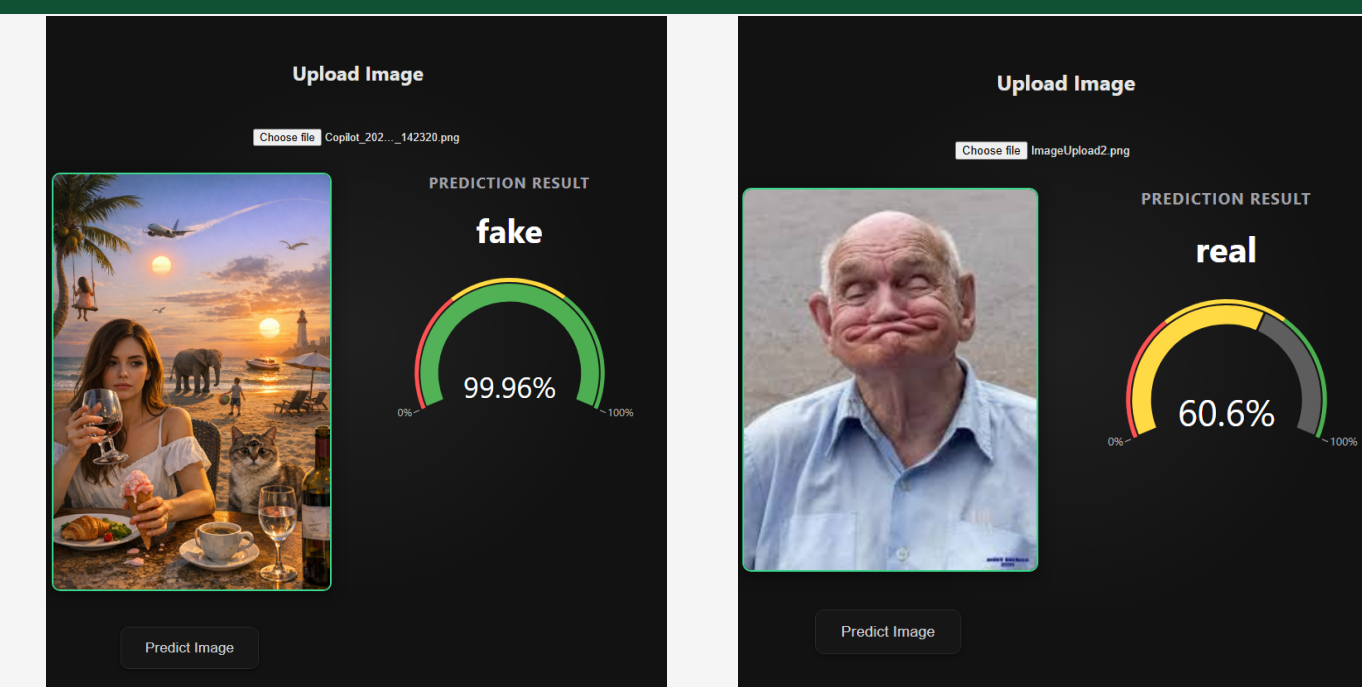


Figure 4: Detected Fake Image

Figure 5: Detected Real Image

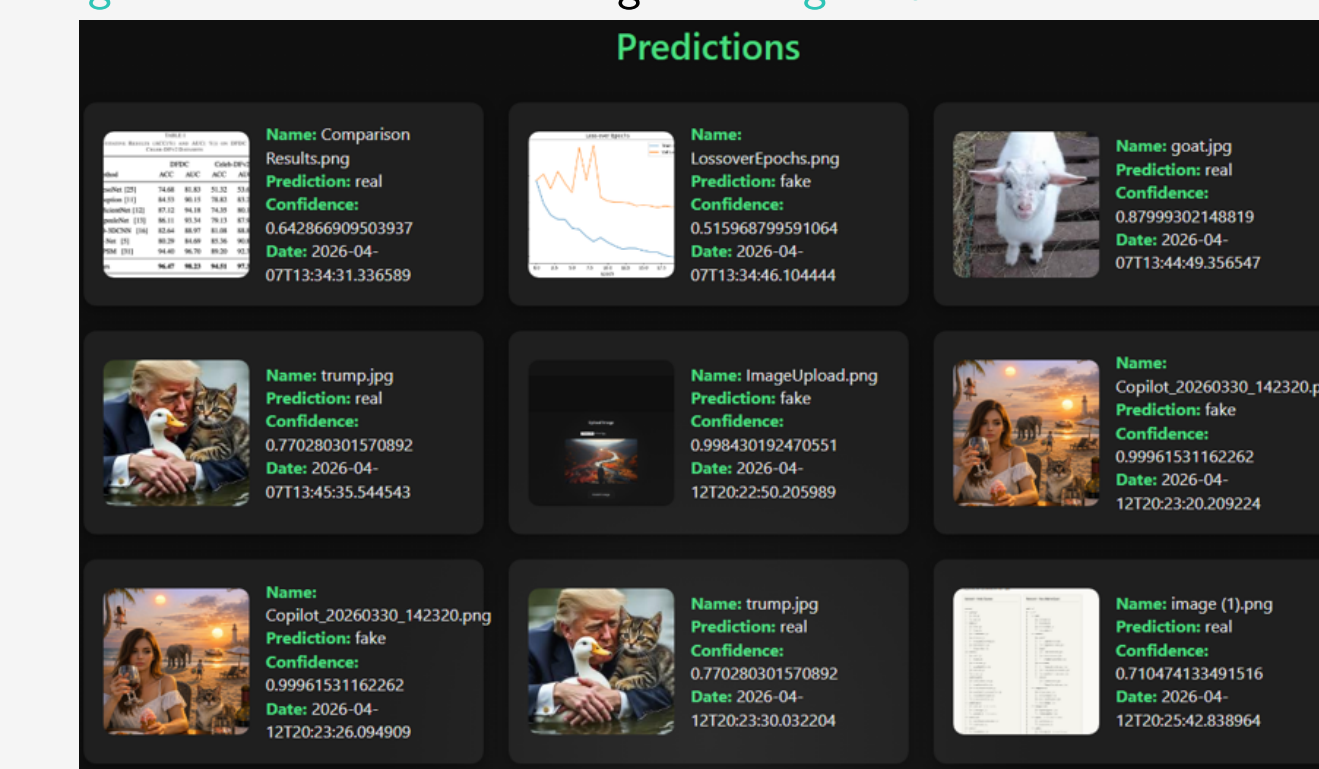


Figure 6: History of Predictions

- Users upload images through the upload page.
- History of uploads shown per user.
- Images stored in AWS S3; predictions stored in SQLite.
- User authentication enables per-user history storage.

Conclusion & Future Work

- PixelProbeB1GI achieves the strongest **cross-dataset generalisation**, performing competitively across ZHANG, CIFAKE, GENImage and GHAI.
- Transfer learning and MHSA integration proved effective.

Future Plans

- Video frame-by-frame detection.
- Mobile app deployment.

References

- MarketingProfs, "Surge of AI-generated images online," 2023. [Online]. Available: <https://www.marketingprofs.com/charts/2023/49890/surge-of-ai-generated-images-online>
- EveryPixel Journal, "AI image statistics: How many images online are generated by AI?," 2023. [Online]. Available: <https://journal.everypixel.com/ai-image-statistics>

Acknowledgements

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Results

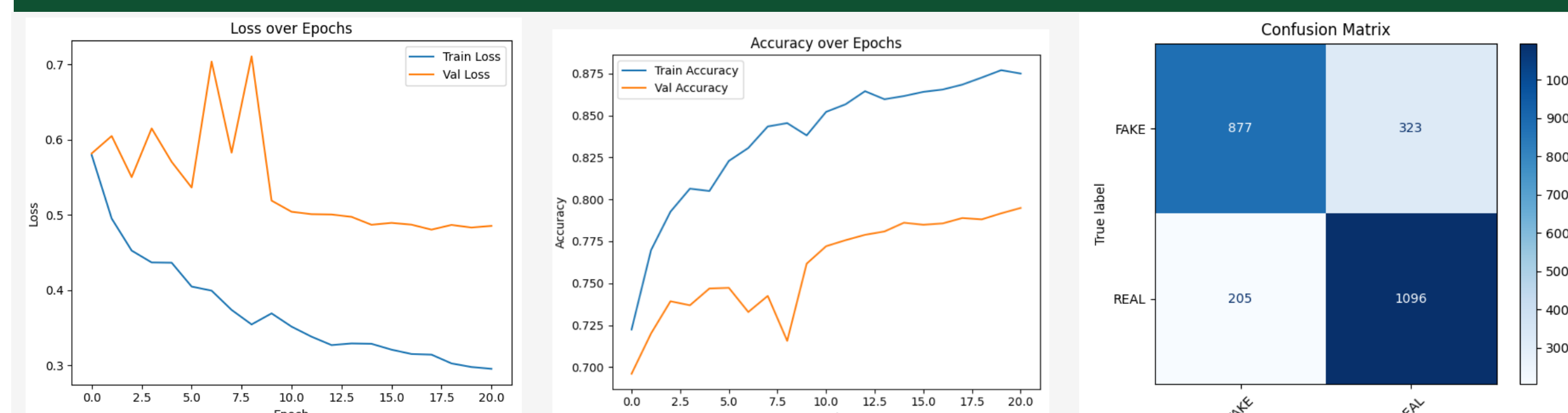


Figure 1: Training Loss (GIV3)

Figure 2: Training Accuracy (GIV3)

Figure 3: Confusion Matrix (GIV3)

Table 1: Image Classification Results

Model	Train Set	ZHANG	CIFAKE	GEN/P365	GHAI
PixelProbeB0	ZHANG	86	50	58	70
PixelProbeB1_V3	ZHANG	94	51	57	78
PixelProbeB1CFV2	CIFAKE	44	95	51	26
PixelProbeB1GI	GenImage	52	61	80	77
PixelProbeB2GI	GenImage	50	56	<u>79</u>	75
PixelProbeB2GIV3	GenImage/Places365	56	54	78	73

Bold = best per column, underline = second best